

Optimization – 1

Extremal problems: numbers, polynomials, functionals

Homework

10 сентября 2026 года

Theoretical minimum. Existence of an extremum on a compact set and why it has to be proved separately for functionals; critical points, boundary points and degenerate points as the three places where an extremum can sit; the alternation property of an extremal polynomial; the three ways of attacking an extremal problem for a functional — pointwise, through extreme points, through a variation — developed in Optimization – 2 and – 3.

1. (Putnam 1998) Find the minimum value of $\frac{(x+1/x)^6 - (x^6+1/x^6) - 2}{(x+1/x)^3 + (x^3+1/x^3)}$ for $x > 0$.
2. (Putnam 1948) What is the maximum of $|z^3 - z + 2|$ over complex z with $|z| = 1$?
3. (Putnam 1984) Find the minimum value of $(u - v)^2 + (\sqrt{2 - u^2} - \frac{9}{v})^2$ for $0 < u < \sqrt{2}$ and $v > 0$.
4. (Putnam 1978) Find the largest A for which there is a polynomial $P(x) = Ax^4 + Bx^3 + Cx^2 + Dx + E$ with real coefficients satisfying $0 \leq P(x) \leq 1$ for $-1 \leq x \leq 1$.
5. (Putnam 1978) Let $p(x) = 2 + 4x + 3x^2 + 5x^3 + 3x^4 + 4x^5 + 2x^6$ and for $0 < k < 5$ put $I_k = \int_0^\infty \frac{x^k}{p(x)} dx$. For which k is I_k smallest?
6. (Putnam 1980) Let C be the class of all real continuously differentiable f on $[0, 1]$ with $f(0) = 0$ and $f(1) = 1$. Determine the largest real number u such that $u \leq \int_0^1 |f'(x) - f(x)| dx$ for all $f \in C$.
7. (Putnam 1991) Find the maximum value of $\int_0^y \sqrt{x^4 + (y - y^2)^2} dx$ for $0 \leq y \leq 1$.

8. (Putnam 2006) For each continuous $f: [0, 1] \rightarrow \mathbb{R}$ put $I(f) = \int_0^1 x^2 f(x) dx$ and $J(f) = \int_0^1 x(f(x))^2 dx$. Find the maximum of $I(f) - J(f)$ over all such f .
9. (Putnam 2011) Suppose f is a function on $[1, 3]$ with $-1 \leq f(x) \leq 1$ for all x and $\int_1^3 f(x) dx = 0$. How large can $\int_1^3 \frac{f(x)}{x} dx$ be?
10. (VJIMC 2013) Let F be the set of all continuous $f: [0, 1] \rightarrow \mathbb{R}$ with $\int_0^x \frac{f(t)}{\sqrt{x-t}} dt \leq 1$ for all $x \in (0, 1]$. Compute $\sup_{f \in F} \int_0^1 f(x) dx$.
11. (VJIMC 2002) Let E be the set of all continuous $u: [0, 1] \rightarrow \mathbb{R}$ satisfying $u^2(t) \leq 1 + 4 \int_0^t s|u(s)| ds$ for all $t \in [0, 1]$, and let $\varphi(u) = \int_0^1 (u^2(x) - u(x)) dx$. Prove that φ attains its maximum on E and find it.